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CS23633 – Cloud Computing Laboratory

Experiment 6: CDN SERVICE SELECTION AND CDN CONFIGURATION.

Aim

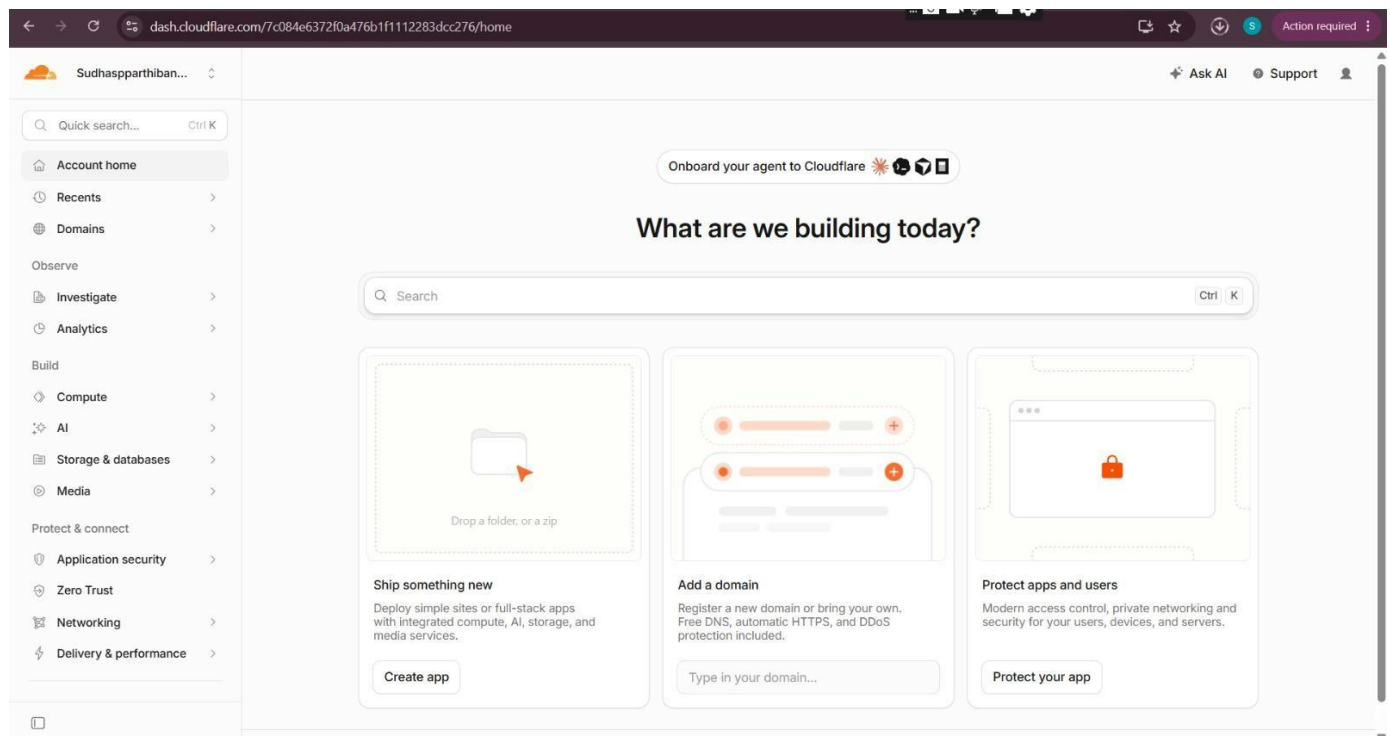
To deploy a simple web application using **Cloudflare Workers**, access it through a public URL, and verify that the content is delivered through the Cloudflare CDN by checking the HTTP response details.

PART A — Creating a Cloudflare Application

Step 1: Open Cloudflare

1. Open a web browser.
2. Visit the **Cloudflare Dashboard**.
3. Sign in to your Cloudflare account.
4. After logging in, open the **Workers & Pages** section.

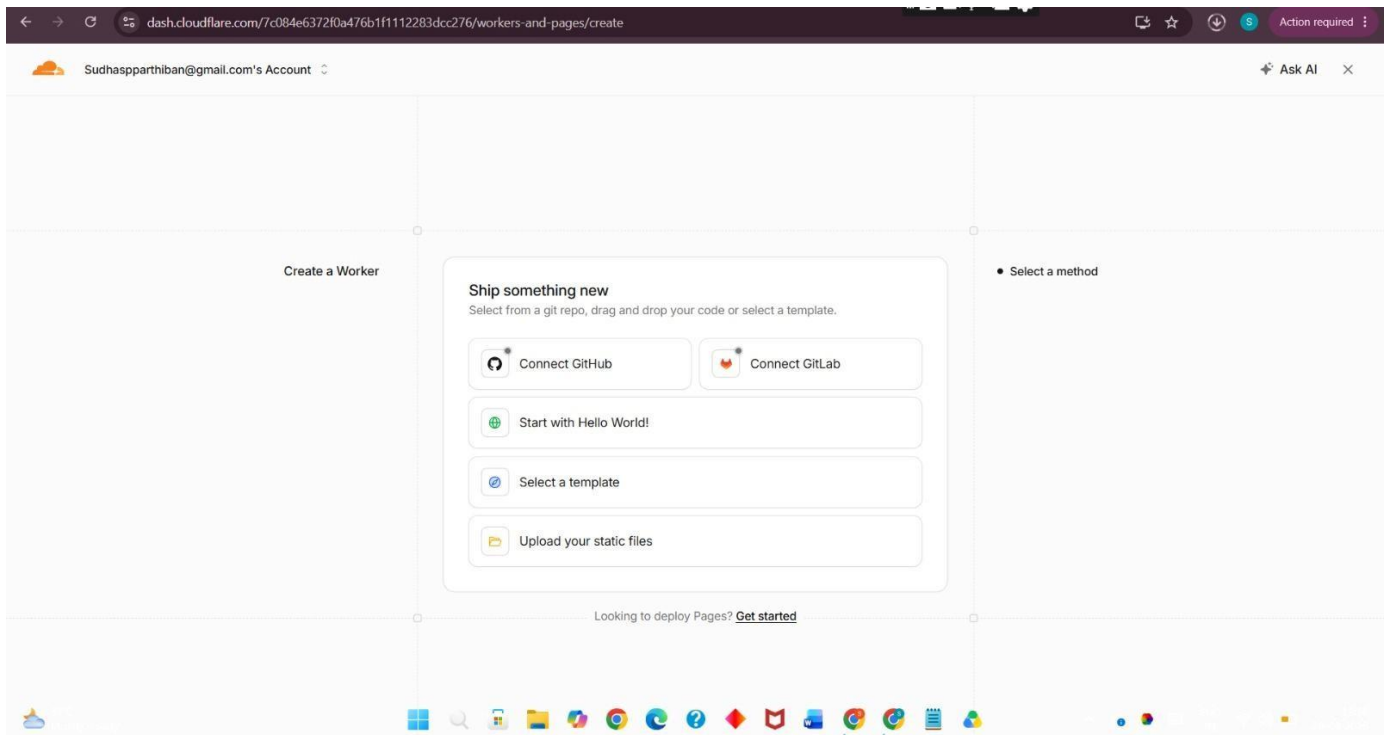
Note: Do not modify any DNS or nameserver settings during this experiment.



Step 2: Create a new application

1. In the Cloudflare dashboard, locate **Ship something new**.
2. Click **Create app**.
3. Proceed with the application creation process.

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Step 3: Choose Hello World

Cloudflare provides different options for creating an application.

Select:

Start with Hello World!

This provides a simple application that can be used for testing the Cloudflare Worker.

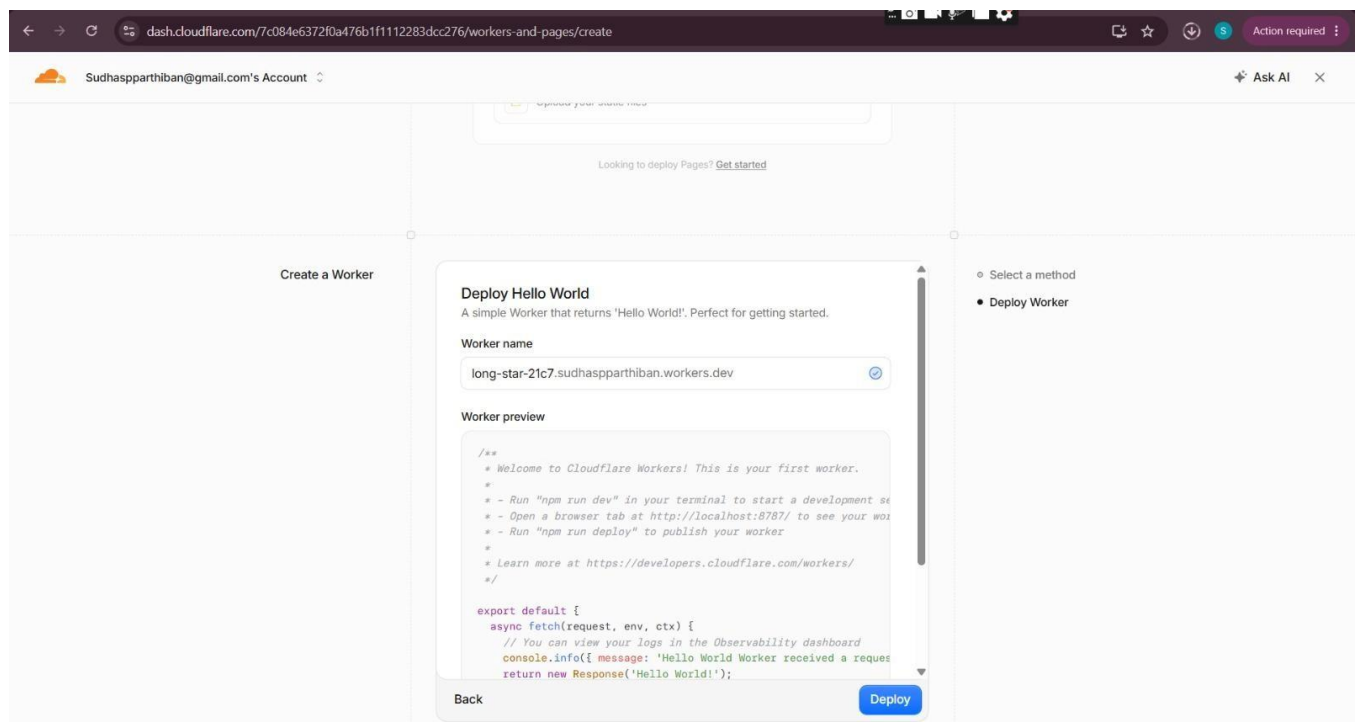
Step 4: Keep the Worker name

Cloudflare automatically generates a name for the Worker.

For example:

long-star-21c7

Do not modify the automatically generated Worker name.



Step 5: Check the default code

The Worker initially contains a simple response such as:

```
return new Response('Hello World!');
```

Keep the default code unchanged at this stage.

Step 6: Deploy the Worker

1. Click the **Deploy** button.
2. Wait until the deployment process is completed.
3. Cloudflare will provide a public URL using the workers.dev domain.

For example:

<https://xxxxxx.workers.dev>

A successful deployment confirms that the Worker has been published.

PART B — Testing the Cloudflare Worker

Step 7: Open the deployed application

1. After deployment, click **Visit** ↗.
2. The Worker will open in a new browser tab.
3. The webpage should display:

Hello World!

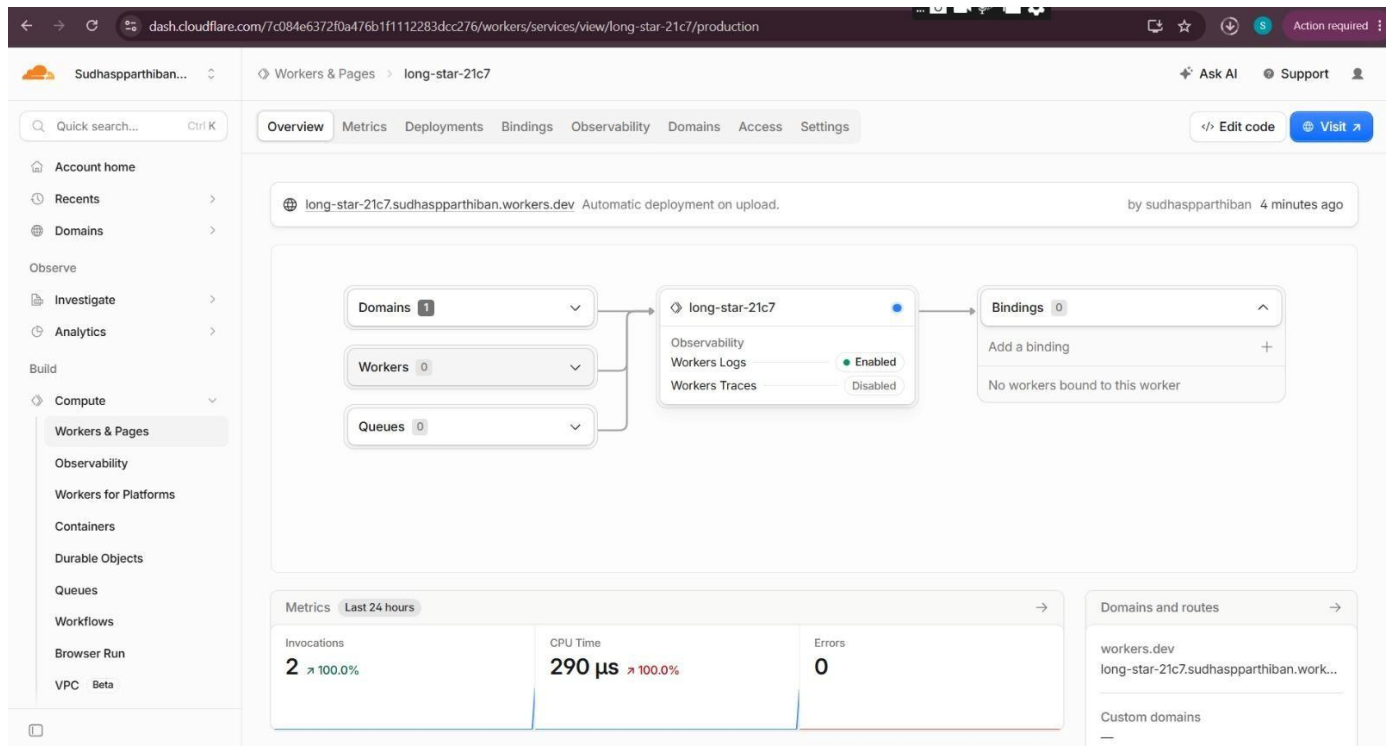
This confirms that the deployed Worker is responding correctly.

Step 8: Check Worker invocation

1. Return to the Cloudflare Worker dashboard.
2. Open the **Metrics** section.
3. Check the number of **Invocations**.

After visiting the Worker URL, the invocation count should increase. This confirms that the deployed Worker has received a request.

To deploy a simple web application using **Cloudflare Workers**, access it through a public URL, and verify that the content is delivered through the Cloudflare CDN by checking the HTTP response details.



PART C — Creating the CDN Experiment Webpage

Step 9: Open the code editor

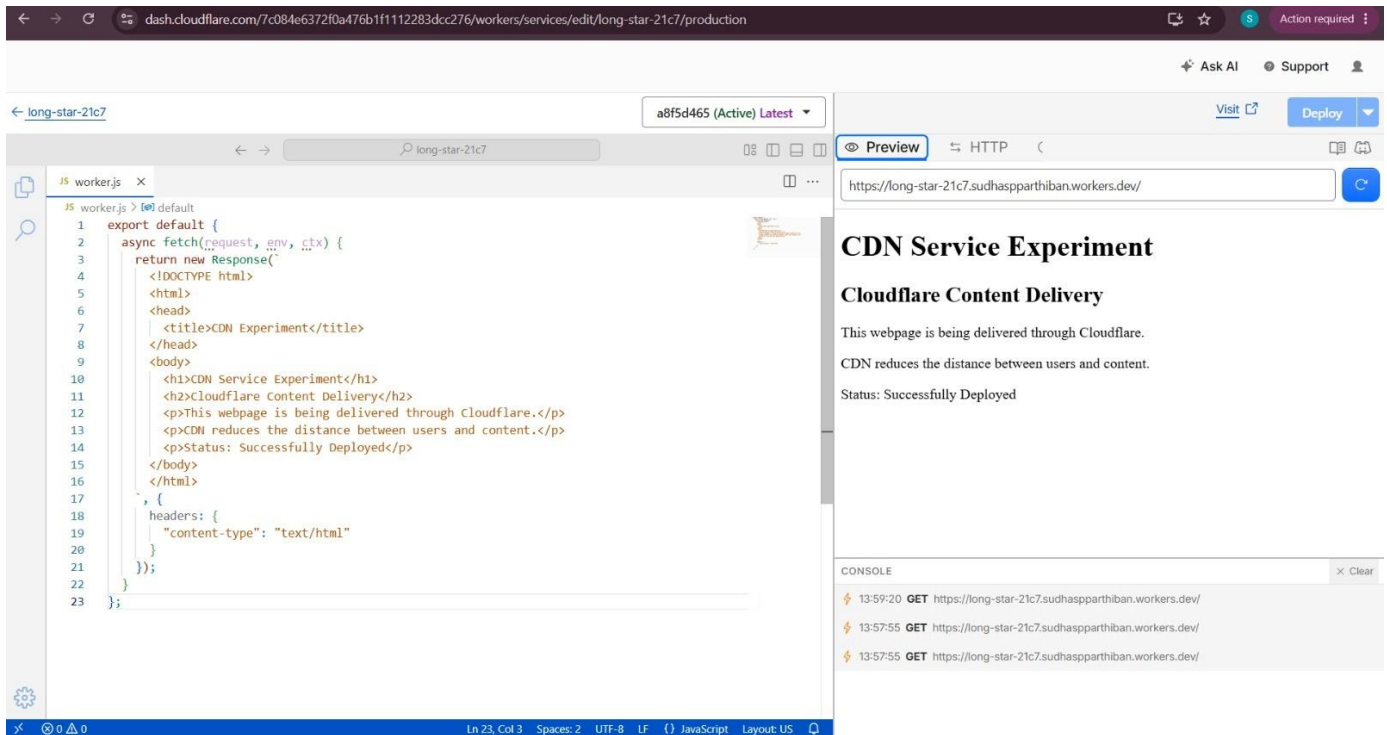
1. Return to the Worker dashboard.
2. Click **Edit code**.
3. Locate the existing Hello World! response.
4. Replace the existing code with the CDN experiment webpage code.

Use the following code:

```
export default {
  async fetch(request, env, ctx) {
    return new Response(`
      <!DOCTYPE html>
      <html>
      <head>
        <title>CDN Experiment</title>
      </head>
      <body>
        <h1>CDN Service Experiment</h1>
        <h2>Cloudflare Content Delivery</h2>
        <p>This webpage is being delivered through Cloudflare.</p>
        <p>CDN reduces the distance between users and content.</p>
        <p>Status: Successfully Deployed</p>
      </body>
    </html>
  `, {
    headers: {
      "content-type": "text/html"
    }
  });
```

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```
}  
};
```



Step 10: Deploy the modified webpage

1. Save the updated code.
2. Click **Deploy**.
3. Wait until Cloudflare completes the deployment.
4. Confirm that the new version has been successfully deployed.

PART D — Verify the CDN Webpage

Step 11: Open the updated webpage

Click **Visit** ↗ again.

The webpage should now display:

CDN Service Experiment

Cloudflare Content Delivery

This webpage is being delivered through Cloudflare.

CDN reduces the distance between users and content.

Status: Successfully Deployed

This confirms that the updated webpage is being served by the Cloudflare Worker.

PART E — Checking the CDN Response

Step 12: Open Developer Tools

1. Keep the deployed webpage open.
2. Press **F12** on the keyboard.
3. Developer Tools will appear.
4. Select the **Network** tab.

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Step 13: Reload the webpage

Press:

Ctrl + R

This reloads the webpage and records the network request.

Step 14: Select the Worker request

In the Network list, locate the request associated with your:
workers.dev

URL.

Click on that request to view its details.

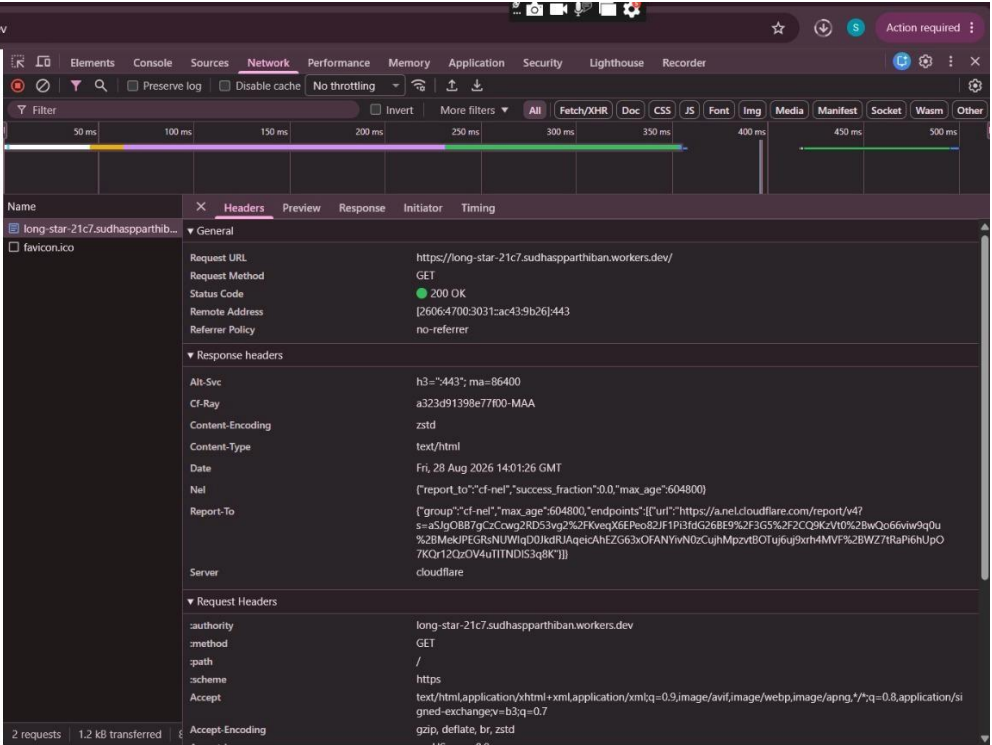
CDN Service Experiment

Cloudflare Content Delivery

This webpage is being delivered through Cloudflare.

CDN reduces the distance between users and content.

Status: Successfully Deployed



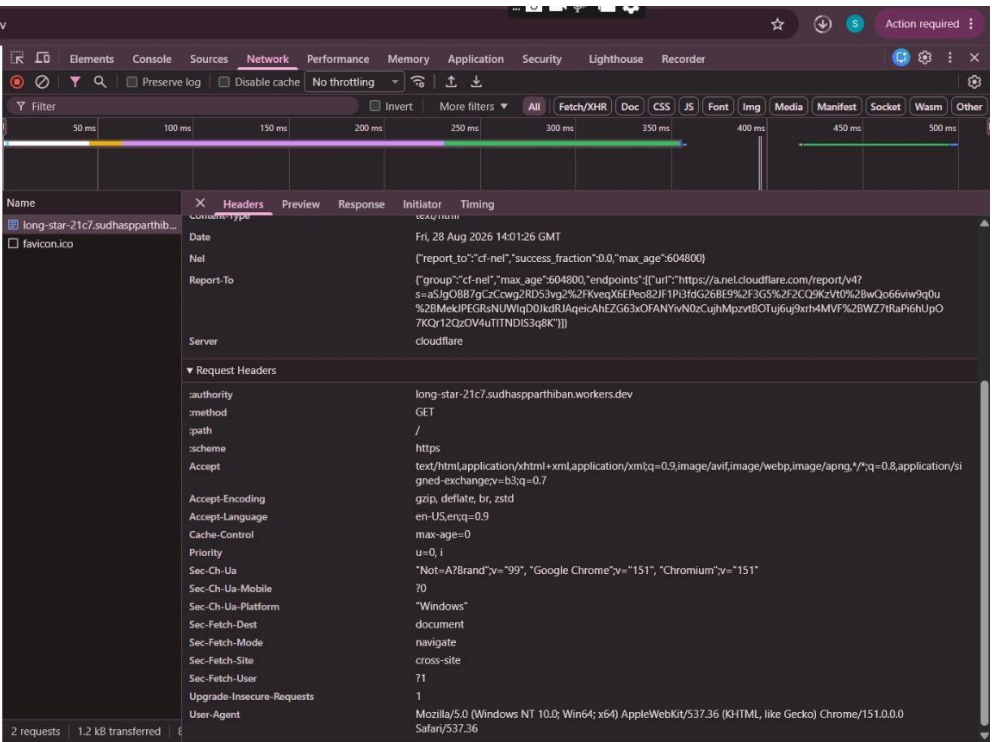
CDN Service Experiment

Cloudflare Content Delivery

This webpage is being delivered through Cloudflare.

CDN reduces the distance between users and content.

Status: Successfully Deployed



Step 15: Check the HTTP headers

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Open the **Headers** section and verify the following information:

Parameter	Expected value
Request Method	GET
Status Code	200 OK
Server	cloudflare
CF-Ray	Cloudflare-generated value
Content-Type	text/html

A 200 OK response indicates that the request was successfully processed. The cloudflare server information and CF-Ray value provide evidence that the request passed through Cloudflare.

RESULT / CONCLUSION

Thus, a web application was successfully deployed using **Cloudflare Workers** and made accessible through a public workers.dev URL. The webpage was successfully delivered through Cloudflare, and the HTTP response was verified using Developer Tools. The **200 OK**, cloudflare server information, CF-Ray, and text/html content type confirmed successful CDN-based content delivery.

Simple Flow

Open Cloudflare → Create Worker → Select Hello World → Deploy → Visit URL → Edit Code → Add CDN webpage → Deploy → Visit webpage → F12 → Network → Reload → Select Worker request → Check Headers → Verify 200 OK